CLINICAL · MOUTH & THROAT

# Oral and oropharyngeal cancer

Two cancers share this chapter and differ in almost everything except their histology. Oral cancer is visible, tied to tobacco, areca nut and alcohol, common in South Asia, and treated first by surgery. Oropharyngeal cancer hides behind the tongue and in the tonsil, presents with a lump in the neck, is increasingly caused by human papillomavirus in younger non-smokers, and is often treated by radiation. Both demand an examination that includes palpation, and both punish delay.

**AT A GLANCE**

- Non-healing ulcer over 2 weeks needs a biopsy.
- HPV has changed oropharyngeal disease.
- Oral cavity: surgery first.
- Treat the neck by depth.
- AJCC 8 stages p16-positive disease separately; cetuximab de-escalation worsened survival.

## 1 CHAPTER ONE Overview

Two regions, one disease.

### 1.1 Subsites

- **Oral cavity:** mucosal lip, buccal mucosa, oral tongue (anterior two-thirds), floor of mouth, upper and lower alveolus (gingiva), retromolar trigone and hard palate.
- **Oropharynx:** base of tongue, tonsil and tonsillar fossa with the pillars, soft palate and uvula (the faucial arch), and the posterior and lateral pharyngeal walls.

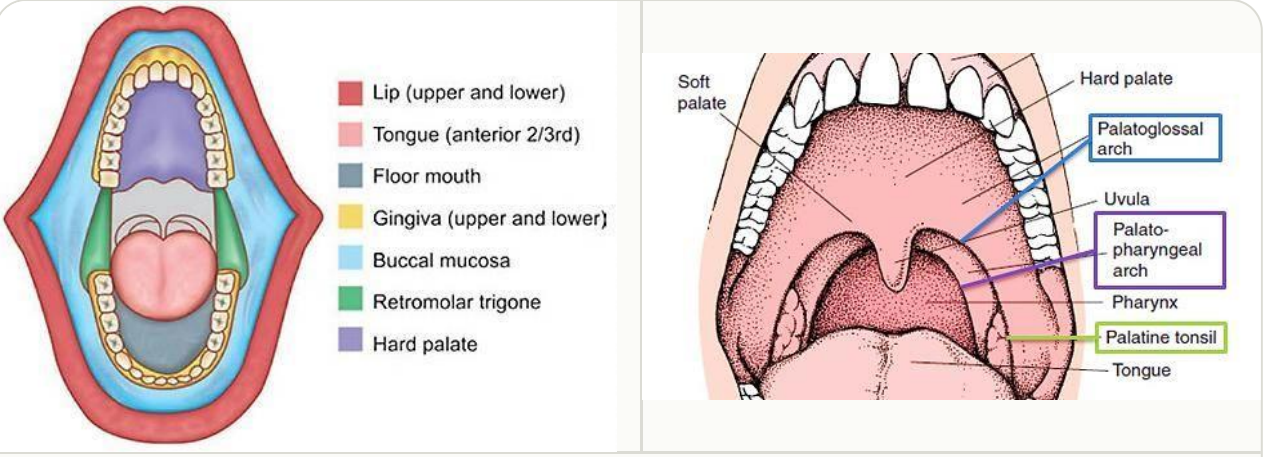



**FIGURE 1** The subsites of the oral cavity, and the oropharynx through the open mouth.
Anatomy atlas · ENT Steps p. 165, ENT Steps p. 210

## 1.2 Epidemiology

India carries one of the highest burdens of oral cancer in the world: lip, tongue and mouth together make up about a fifth of all male cancers there, with buccal mucosa and tongue the leading sites, driven by tobacco chewing with lime, pan and areca nut, bidi smoking and alcohol. In the West the lip is the commonest oral subsite and tongue cancer is rising in young adults. Oropharyngeal cancer is rising rapidly in Europe and North America because of HPV.

## 1.3 Causes

- **Tobacco:** smoking increases oral cancer about sixfold; chewing tobacco places carcinogen at the site where the quid is held; *reverse smoking* (the burning end inside the mouth) causes palatal cancer.
- **Alcohol:** multiplies, rather than adds to, the effect of tobacco.
- **Areca nut and pan:** directly and through submucous fibrosis.
- **Human papillomavirus (HPV 16):** most cancers of the tonsil and base of tongue in North America; associated with oral sexual behaviour rather than tobacco.
- **Others:** chronic trauma from jagged teeth and dentures, poor oral hygiene, Plummer–Vinson syndrome, immunosuppression, sunlight (lower lip), and rare syndromes (Fanconi anaemia).

### 1.4 Field cancerisation

Tobacco and alcohol alter the whole mucosa of the upper aerodigestive tract, so dysplasia and cancer arise in more than one place. About 15% of patients have, or will develop, a second primary in the head and neck, oesophagus or lung.

### 1.5 Pathology

Squamous cell carcinoma in about 90%; minor salivary gland carcinomas (adenoid cystic, mucoepidermoid, adenocarcinoma), lymphoma (especially tonsil and base of tongue), melanoma (palate and gum) and Kaposi sarcoma make up the rest. Verrucous carcinoma is a slow, warty, well-differentiated variant of the buccal mucosa that rarely metastasises.

## 2 CHAPTER TWO Clinical picture

Look, feel, and feel the neck.

### 2.1 Gross forms

Exophytic, ulcerative (a non-healing ulcer with raised, rolled, everted edges, a shaggy base and induration), infiltrative, and superficially spreading (often arising in leukoplakia or erythroplakia).

### 2.2 Symptoms

- An ulcer or lump that does not heal; bleeding.
- Pain, often *referred to the ear* through the lingual and auriculotemporal nerves (V3) or the glossopharyngeal nerve.
- Difficulty in chewing, swallowing and speaking; a *hot-potato voice* with tongue base tumours; loosening of teeth or ill-fitting dentures; trismus when the pterygoids or masseter are invaded; numbness of the lip from inferior alveolar nerve invasion.
- A lump in the neck, sometimes the only complaint.
- Foul breath, weight loss and, late, a fungating mass or orocutaneous fistula.

**REFERRAL NOW**

A mouth ulcer or white or red patch that persists beyond three weeks, an unexplained lump in the mouth or neck, unexplained tooth mobility, persistent one-sided sore throat or earache with a normal ear: refer urgently for examination and biopsy.

### 2.3 Examination

Inspect every subsite with good light, palpate the lesion for induration and depth, palpate bimanually the floor of mouth and tongue, examine the tongue base and tonsils by palpation and endoscopy, and examine both sides of the neck level by level. Cranial nerves V, VII and XII; mouth opening; teeth.

## 3 CHAPTER THREE Oral cavity cancer

Site by site.

### 3.1 Lip

The commonest oral subsite in the West; more than 95% squamous, 95% on the *lower lip*, between the midline and the commissure, in older men with outdoor occupations; basal cell carcinoma is relatively commoner on the upper lip. Early blistering, crusting or ulceration on a background of actinic cheilitis. Nodes (submental and submandibular, bilateral near the midline) are late and occur in only about 10%.



**FIGURE 2** Carcinoma of the upper lip and oral commissure, with leukoplakia of the buccal mucosa.

Dhingra, fig. 44.10

### 3.1.1 Treatment

Excision with a safe margin. Defects up to a third of the lip close primarily after a V or W excision; a third to a half with lip-switch flaps (Abbe, or Estlander at the commissure); a half to two-thirds with a Karapandzic flap; larger defects with Gillies fan, Bernard–Burow or free flaps, always realigning the vermilion border. Radiotherapy is an alternative for small lesions; neck dissection for nodes. Cure exceeds 90% in early disease; upper lip and commissure tumours do worse.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Surgery for cancer of the lip.

**Cancer of the Lip**

## 3.2 Buccal mucosa

Very common in India, equal in both sexes, at the angle of the mouth, along the occlusal line or in the gingivobuccal sulcus where the quid is kept; often multicentric because the whole cheek is "condemned". Exophytic or ulceroinfiltrative; it spreads deeply through buccinator and fat to the skin, back to the retromolar trigone and pterygoids (trismus), and up and down into the maxilla

and mandible. About half have nodes (submandibular, then upper jugular). Verrucous carcinoma favours this site.



**FIGURE 3** Carcinoma of the buccal mucosa.
Dhingra, fig. 44.11

#### 3.2.1 Treatment

T1 lesions by excision. Larger lesions, and any with muscle or bone involvement, by wide excision with marginal or segmental mandibulectomy or partial maxillectomy, reconstruction with local, pectoralis major or free flaps, neck dissection, and postoperative radiotherapy for stage III and IV.

## 3.3 Oral tongue

Usually men of fifty to seventy, though increasingly young adults; may arise in leukoplakia, at a chronic dental ulcer, or in smokers and chewers. The commonest site is the *middle of the lateral border* and the ventral surface; the tip and dorsum less often. It appears as an exophytic papillary growth, an indurated ulcer with rolled edges, or a submucosal indurated nodule. It infiltrates the tongue musculature (fixation, slurred speech), the floor of mouth and mandible. Nodes: the lateral tongue drains to the submandibular and upper jugular nodes (and can skip to level IV), the tip to the submental and jugulo-omohyoid nodes; midline and ventral lesions spread to both sides.

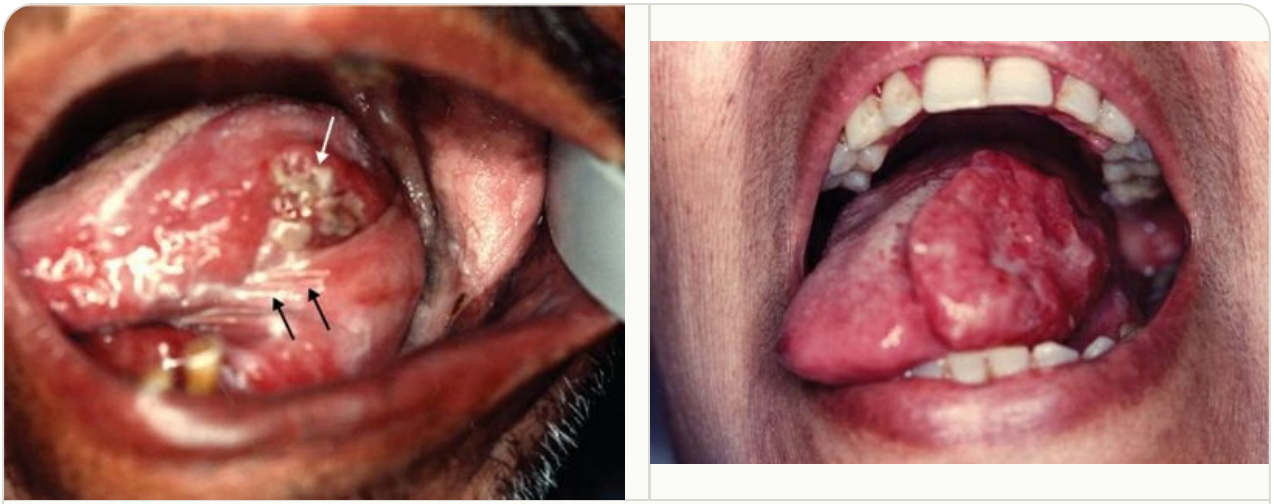


**FIGURE 4** First, carcinoma of the lateral border of the tongue (arrow) with leukoplakia of the floor of mouth (double arrows). Second, an ulcerative squamous carcinoma of the tongue in a 40-year-old woman.

Dhingra, figs 44.12 and 44.13

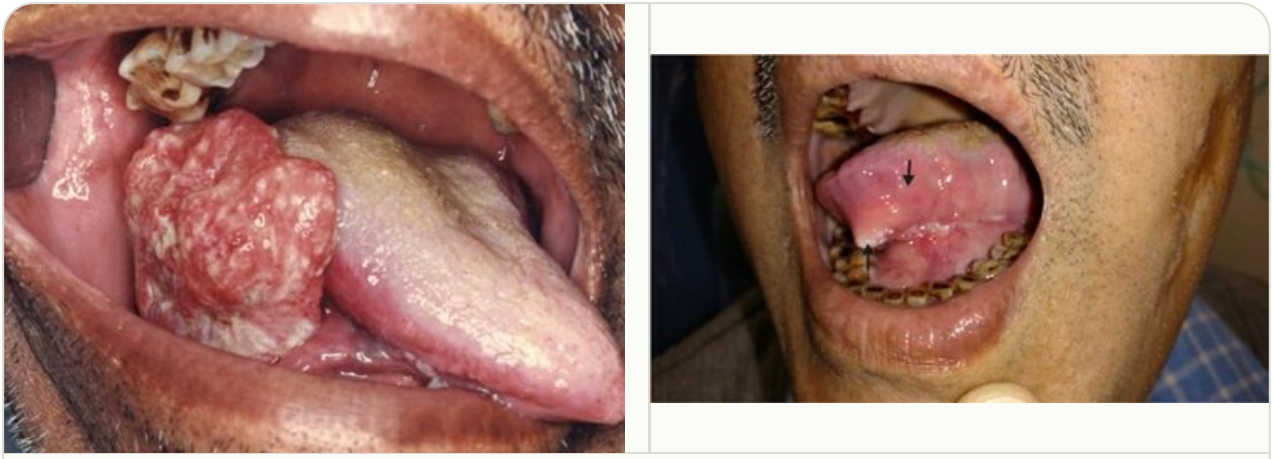


**FIGURE 5** First, an exophytic growth on the right lateral border of the tongue. Second, cancer of the undersurface of the tongue.

Dhingra, figs 44.14 and 44.15

**DEPTH OF INVASION**

The thickness of a tongue cancer predicts nodal spread better than its width: occult metastases rise sharply beyond about 4 mm. Current staging (AJCC 8th edition) therefore includes depth of invasion in the T category, and elective neck dissection is advised for tumours deeper than about 3 to 4 mm.

### 3.3.1 Treatment

T1 and T2: partial glossectomy (transoral or with a pull-through), healing by secondary intention, primary closure, graft or free flap; radiotherapy (external beam with brachytherapy) is an alternative. Elective selective neck dissection of

> levels I to III (supraomohyoid) for an N0 neck when depth exceeds the threshold, bilateral for midline lesions; sentinel node biopsy is an alternative in early cancers. Advanced lesions: hemiglossectomy or total glossectomy with the floor of mouth, mandibulectomy and neck dissection (the *commando operation*), free flap reconstruction and postoperative radiotherapy or chemoradiation; total glossectomy may require laryngectomy to prevent aspiration.

## 3.4 Floor of mouth

Squamous carcinoma, men four times more than women, usually starting anteriorly near the submandibular duct openings, which may obstruct and enlarge the gland. Ulcerative or infiltrative, spreading to the ventral tongue, lingual gingiva, mandible and submental space; a submandibular swelling may be gland or node. Small lesions by excision or radiotherapy; larger ones by wide excision with marginal or segmental mandibulectomy; elective neck treatment even for N0 stage II disease, since occult nodes occur in about 40%; nodes often bilateral.



**FIGURE 6** Ulcerative squamous carcinoma of the floor of mouth (arrows) in a 55-year-old man. Dhingra, fig. 44.16

## 3.5 Alveolus (gingiva)

Mostly men, on the lower jaw behind the first molar; early invasion of the bone and along the inferior alveolar nerve; nodes to the submandibular and upper jugular

groups. Surgery is preferred because radiotherapy risks osteoradionecrosis: marginal mandibulectomy for superficial lesions, segmental or hemimandibulectomy when the medulla is invaded; partial maxillectomy for upper alveolar lesions.

## 3.6 Retromolar trigone

Primary, or by extension from the gingiva, floor of mouth, buccal mucosa or faucial arch; it reaches the pterygomandibular raphe, pterygoids (trismus) and mandible early. Wide excision with mandibular resection and neck dissection.



**FIGURE 7** Cancer of the right retromolar trigone with trismus in a 36-year-old man. Dhingra, fig. 44.17

## 3.7 Hard palate

Squamous or, more often, minor salivary gland carcinoma (adenoid cystic, mucoepidermoid, adenocarcinoma); squamous cancer is linked to reverse smoking. A painless irregularity felt with the tongue, a superficial ulcer, or a submucosal mass; it may invade the nasal floor and antrum and must be told apart from an antral carcinoma breaking through. Resection with the underlying bone or a partial maxillectomy, closed with an obturator prosthesis or flap.



**FIGURE 8** A mixed salivary tumour of the palate.
Dhingra, fig. 44.18

**OPERATIVE GUIDES IN THEATRE** 6 operations

Glossectomy, floor of mouth excision, transoral inferior maxillectomy and mandibulectomy.

**Partial Glossectomy**
Partial glossectomy
**Total Glossectomy**
Total glossectomy
**Excision of Cancer of the Floor of the Mouth**
Floor of mouth
**Transoral Inferior Maxillectomy**
**Marginal Mandibulectomy**
**Segmental Mandibulectomy**

## 3.8 The mandible

Tumour reaches the mandible through the alveolar crest in the dentate jaw and through the occlusal surface in the edentulous jaw. Periosteal involvement needs at least a *marginal (rim) mandibulectomy*, preserving the lower border; invasion of the medulla, or an edentulous atrophic mandible, needs a *segmental* resection, reconstructed with a fibula free flap. Radiotherapy to a mandible already invaded leads to osteoradionecrosis.

# 4 CHAPTER FOUR Oropharyngeal cancer

Behind the tongue.

## 4.1 HPV and the changing disease

HPV-POSITIVE AND HPV-NEGATIVE OROPHARYNGEAL CANCER

| FEATURE | HPV-POSITIVE | HPV-NEGATIVE |
|---|---|---|
| Typical patient | Younger, healthier, male (about 9:1), non-smoker | Older, smoker and drinker |
| Cause | HPV 16; oral sexual exposure | Tobacco and alcohol |
| Marker | p16 overexpression on immunohistochemistry | p16 negative |
| Primary | Often small, in the tonsil or tongue base crypts | Often larger, infiltrative |
| Nodes | Early, often large and *cystic* | Solid |
| Prognosis | Much better; staged separately (AJCC 8th) | Worse |

The viral E6 and E7 oncoproteins inactivate p53 and Rb, and p16 is overexpressed as a result. Vaccination against HPV is expected to prevent most of these cancers.

**THE CYSTIC NECK LUMP IN AN ADULT**

A cystic lump in the upper neck of an adult is a metastasis from an HPV-related tonsil or tongue base cancer until proven otherwise, not a branchial cyst. Examine and image the oropharynx, aspirate for p16 or HPV, and if no primary is seen, perform tonsillectomy and tongue base biopsy or mucosectomy.

## 4.2 Base of tongue

Silent for a long time; early symptoms (soreness, a lump feeling, mild dysphagia) are attributed to the lingual tonsil. Many present with nodes, often bilateral: about 70% have neck metastases at presentation, so the tongue base is a classic site of the occult primary. Late: referred otalgia, dysphagia, bleeding, a hot-potato voice and tongue fixation. Deeply infiltrative into the tongue, vallecula, pre-epiglottic space,

epiglottis and tonsil. It is larger on palpation than it looks in the mirror: always palpate, preferably under anaesthesia.

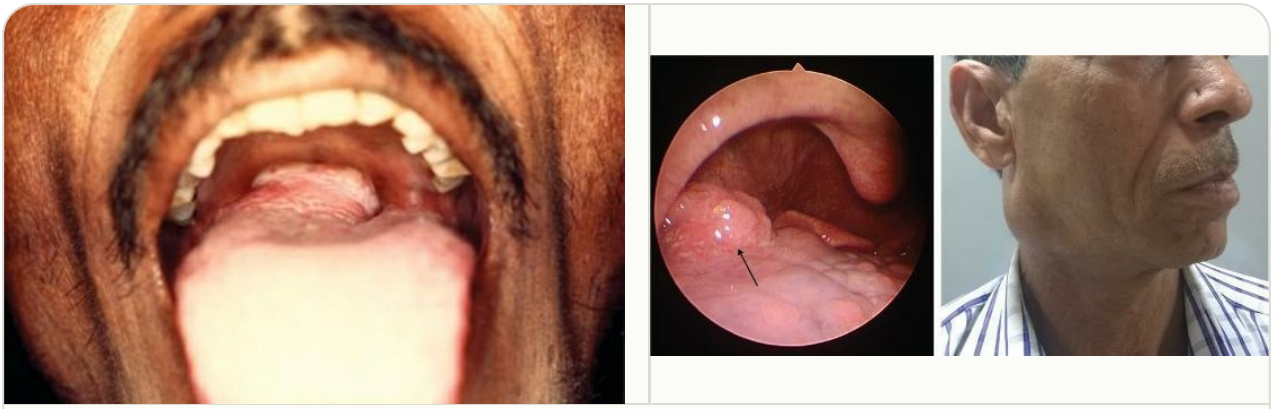


**FIGURE 9** First, an exophytic growth at the base of the tongue. Second, carcinoma of the right base of tongue: (A) endoscopic view, (B) secondaries in the jugulodigastric nodes.

Dhingra, figs 53.1 and 53.2

## 4.3 Tonsil and tonsillar fossa

Squamous carcinoma presents as an ulcer with a necrotic base, or as a smooth, enlarged tonsil when it lies within the crypts; lymphoma as a smooth, one-sided enlargement, sometimes ulcerated. Persistent sore throat, dysphagia, otalgia and a neck lump, later bleeding, fetor and trismus. It spreads to the palate, pillars, tongue base, pharyngeal wall, pterygoids and mandible, and the parapharyngeal space; about half have nodes at presentation, the jugulodigastric node first.

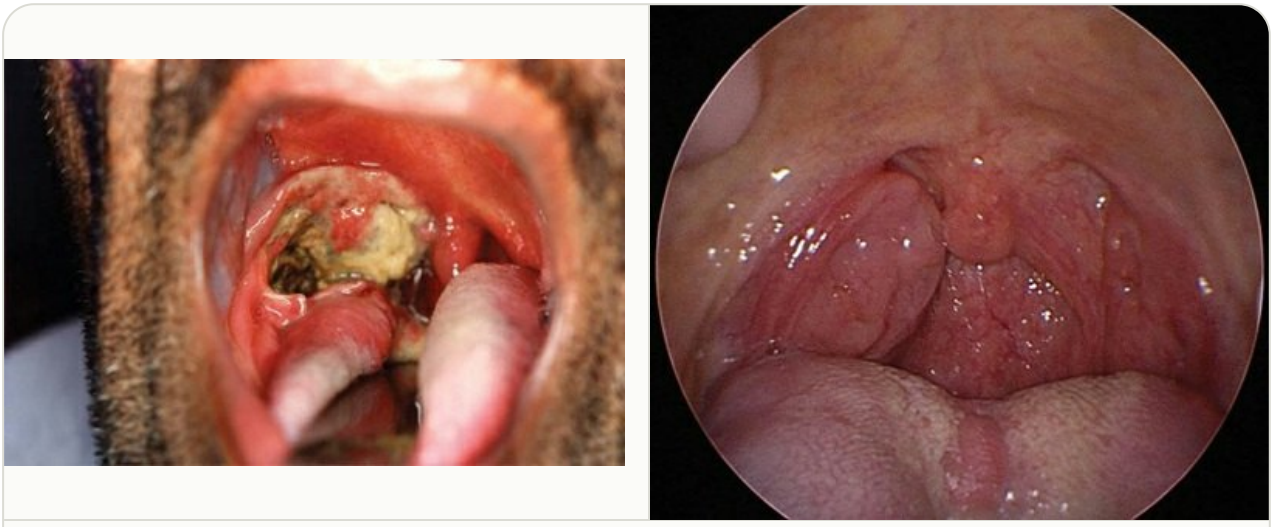


**FIGURE 10** First, squamous cell carcinoma involving the tonsil, pillars and soft palate. Second, lymphoma of the right tonsil presenting as one-sided tonsillar enlargement.

Dhingra, figs 53.3 and 53.4

### THE ONE-SIDED ENLARGED TONSIL

| CAUSE | CLUE |
|---|---|
| Normal asymmetry or a medially placed tonsil | Long-standing, smooth, no symptoms |
| Unilateral tonsillitis, peritonsillitis, quinsy | Pain, fever, trismus |
| Tonsillolith, cyst | Gritty or yellow |
| Parapharyngeal tumour | Tonsil pushed medially, normal mucosa |
| Carcinoma or lymphoma | Progressive, ulcerated, firm, nodes, otalgia; biopsy or tonsillectomy |

## 4.4 Faucial arch (soft palate, uvula, anterior pillar)

Usually well-differentiated, superficially spreading squamous carcinoma on the oral surface, noticed early and slow to metastasise, behaving more like oral cancer; often accompanied by other primaries. Radiotherapy or transoral excision.



**FIGURE 11** An ulcerative lesion of the palatine arch: a well-differentiated squamous cell carcinoma.

Dhingra, fig. 53.5

### 4.5 Posterior and lateral pharyngeal walls

Silent, spreading submucosally up to the nasopharynx and down to the hypopharynx, and back to the prevertebral fascia; nodes in about 60%, often bilateral and retropharyngeal. Radiotherapy or excision through a lateral pharyngotomy, often with mandibulotomy, and neck dissection.

### 4.6 Benign tumours of the oropharynx

Papilloma (pedunculated, on the tonsil, palate or pillars; excised), haemangioma (treated only if growing or bleeding), pleomorphic adenoma of the palate (excised completely), vallecular mucous cysts (excised or marsupialised), and rarely lipoma, fibroma and neuroma. Parapharyngeal tumours (deep-lobe parotid, schwannoma, paraganglioma, lipoma, carotid aneurysm) push the tonsil and palate medially and mimic oropharyngeal tumours: never biopsy through the mouth without imaging.

## 5 CHAPTER FIVE Evaluation and staging

Before treatment.

### 5.1 Investigations

- **Biopsy** of the primary; fine-needle aspiration of nodes; p16 immunohistochemistry (and HPV testing) for every oropharyngeal cancer.
- **CT and MRI** of the primary and neck: depth, bone invasion (CT and orthopantomogram for the mandible), extension into the pterygoids, parapharyngeal space, pre-epiglottic space and skull base; nodes, including retropharyngeal.
- **Chest CT or PET-CT** for distant metastases and second primaries; PET-CT also for the unknown primary.
- **Panendoscopy** under anaesthesia (laryngoscopy, pharyngoscopy, oesophagoscopy) to map the tumour and look for synchronous cancers.
- Dental assessment before radiotherapy; nutrition and swallowing assessment.

## 5.2 TNM essentials

T STAGING (AJCC 8TH EDITION)

| CATEGORY | ORAL CAVITY | OROPHARYNX |
|---|---|---|
| T1 | 2 cm or less, depth of invasion 5 mm or less | 2 cm or less |
| T2 | 2 cm or less with depth over 5 mm, or 2 to 4 cm with depth 10 mm or less | Over 2 cm, up to 4 cm |
| T3 | Over 4 cm, or depth over 10 mm | Over 4 cm, or extension to the lingual surface of the epiglottis |
| T4a | Cortical bone, maxillary sinus, skin of face (and, for lip, inferior alveolar nerve or floor of mouth) | Larynx, extrinsic tongue muscle, medial pterygoid, hard palate or mandible |
| T4b | Masticator space, pterygoid plates, skull base, or encasing the internal carotid | Lateral pterygoid, pterygoid plates, lateral nasopharynx, skull base, or encasing the carotid |

For oral cavity and HPV-negative oropharyngeal cancer, N staging follows node size, number and side, and *extranodal extension* (which upstages); for HPV-positive oropharyngeal cancer the N categories are simpler and the stages lower, reflecting the better prognosis. Dhingra prints the older (6th edition) tables, based on size alone.

# 6 CHAPTER SIX Treatment

By site, stage and the person.

## 6.1 Principles

Treatment is planned by a multidisciplinary team (surgeons, radiation and medical oncologists, radiologists, pathologists, reconstructive surgeons, dentists, speech and swallowing therapists, dietitians), balancing cure against speech, swallowing and appearance.

- **Oral cavity:** surgery first. Early (stage I and II): excision with a margin, with elective neck treatment according to depth. Advanced (III and IV): resection, reconstruction and neck dissection, followed by radiotherapy, or chemoradiation when margins are positive or there is extranodal extension.
- **Oropharynx:** early: radiotherapy (IMRT) or transoral surgery (robotic or laser) with neck dissection, both treating the neck. Advanced: concurrent chemoradiation with cisplatin (organ preservation), or surgery with adjuvant (chemo)radiation; surgery through a mandibulotomy is mostly reserved for salvage. Radiosensitive lymphoepithelioma and lymphoma are treated by radiotherapy or chemotherapy.
- **Systemic agents:** cisplatin is the standard radiosensitiser; cetuximab (an EGFR antibody) is an alternative for those unfit for cisplatin, though inferior in HPV-positive disease; immunotherapy for recurrent and metastatic disease.

## 6.2 The neck

- **N0:** elective selective neck dissection (levels I to III for oral cavity, II to IV for oropharynx) or elective radiotherapy when the risk of occult metastasis exceeds about 20%; sentinel node biopsy in early oral cancer.
- **N+:** comprehensive (modified radical) neck dissection, followed by radiotherapy for multiple nodes or extranodal extension.

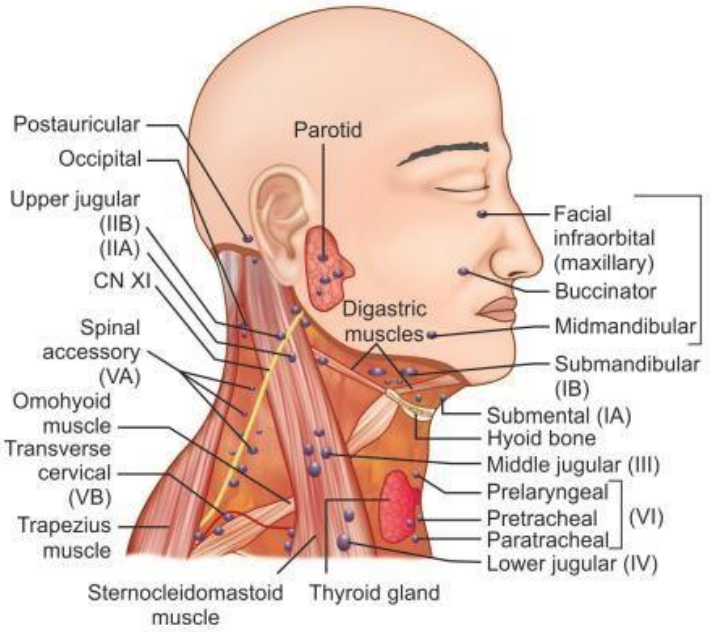



**Fig. 6:** Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

**FIGURE 12** The lymph node levels of the neck.

Anatomy atlas · ENT Steps p. 285

## 6.3 Reconstruction

Primary closure, skin grafts, local flaps (nasolabial, tongue, facial artery musculomucosal), regional flaps (pectoralis major, deltopectoral) and free flaps (radial forearm for the tongue and floor, anterolateral thigh for bulk, fibula for the mandible). Tracheostomy protects the airway after major resections.

**OPERATIVE GUIDES IN THEATRE** 7 operations

Glossectomy, floor of mouth and tonsil surgery, base of tongue resection, neck dissection and free flaps.

**Partial Glossectomy**
Partial glossectomy

**Total Glossectomy**
Total glossectomy

**Excision of Cancer of the Floor of the Mouth**
Floor of mouth

**Transoral and Robotic Tonsil Surgery for Cancers of the Base of the Tongue and Pharynx**
Tonsil surgery

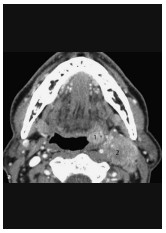
**Transoral Robotic Surgery for Neoplasms of the Base of Tongue**
Base of tongue

**Selective Neck Dissection**
Neck dissection

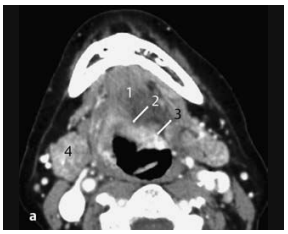
**Microvascular Reconstruction of the Head and Neck**
Free flaps

IMAGING IN THE APP 4 cases

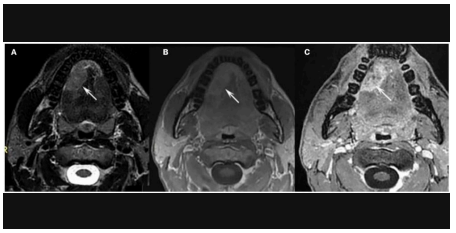
Tonsillar carcinoma, base of tongue tumour, oral and oropharyngeal squamous carcinoma, and synchronous malignancies.



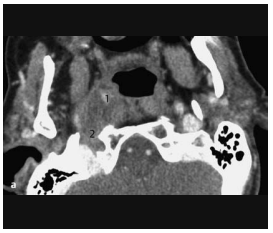
**Tonsillar Carcinoma**
Tumours · 8 images



**Oropharyngeal Tumor, Base of Tongue**
Tumours · 11 images



**Squamous cell carcinoma: oral cavity, oropharynx and larynx**
After surgery · 14 images



**Multiple Synchronous Malignancies**
Tumours · 3 images

## 6.4 Palliation

Radiotherapy or chemotherapy for pain and bleeding, tracheostomy and gastrostomy for airway and nutrition, and good analgesia and palliative care.

## 6.5 Prognosis

Early oral cancer is cured in most patients; advanced disease in roughly a third to a half. Extranodal extension, nodes low in the neck, positive margins and perineural invasion worsen survival. HPV-positive oropharyngeal cancer has markedly better survival than HPV-negative. Lifelong follow-up looks for recurrence (most in the first two years) and second primaries, and stopping smoking improves outcome.

## 6.6 Chemoprevention

Retinoids, beta-carotene and vitamin E can induce regression of leukoplakia and reduce second primaries while taken, but no agent has replaced stopping tobacco, areca nut and alcohol.

# 7 CHAPTER SEVEN
# Non-squamous cancers

The others.

- **Minor salivary gland carcinoma:** 80 to 90% of minor salivary tumours are malignant in some series; the palate is the commonest site; adenoid cystic (about 40%, with perineural spread), adenocarcinoma and mucoepidermoid. Wide excision with neck dissection for nodes, and radiotherapy.
- **Mucosal melanoma:** palate and gum, sixth decade, men twice as often; pigmented, later ulcerated, sometimes amelanotic; wide excision including bone; five-year survival around 15%.
- **Lymphoma:** mostly non-Hodgkin, in the tonsil and tongue base, as a smooth bulky submucosal mass, with neck nodes in many; chemotherapy and radiotherapy.
- **Kaposi sarcoma:** reddish-purple palatal plaques, mainly in AIDS; responds to antiretroviral treatment, radiotherapy or chemotherapy.

### HPV, STAGING AND DE-ESCALATION (OXFORD TEXTBOOK)

HPV-positive oropharyngeal cancer, mostly HPV-16, is now a distinct disease with a better prognosis. Smoking worsens prognosis whatever the HPV status. Local and regional failure is less common in HPV-positive disease, but distant metastases, mostly to the lung, occur at similar rates, so chest surveillance still matters. The **AJCC eighth edition** stages p16-positive and p16-negative tumours separately, with a separate pathological staging after surgery, so that stage groups reflect the better outlook; but guidelines do not yet change treatment on HPV status alone. Neck aspirates and biopsies are tested for p16, and dental extractions needed before radiotherapy are best done at the staging anaesthetic to reduce the risk of osteoradionecrosis.

**Treatment.** Transoral surgery (laser or robotic) with neck dissection as the only treatment is appropriate for T1–T2 tumours with no nodes or a single node, when margins are clear and there are no adverse pathological features; patients must accept that adjuvant treatment may be needed. Transoral resection with adjuvant therapy gives similar control and survival to chemoradiotherapy, so expected morbidity and function guide the choice. Chemotherapy is added to postoperative radiotherapy for extranodal extension or positive margins, and patients with gross extranodal extension on imaging are better treated with primary chemoradiotherapy to avoid triple therapy. **De-escalation** (lower radiation dose, surgery alone, or replacing cisplatin) aims to reduce long-term toxicity in HPV-positive patients; the De-ESCALaTE and RTOG 1016 trials showed that replacing cisplatin with cetuximab *worsened* survival, so de-escalation belongs only within trials.

# 8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

## 8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Risk factors,** field cancerisation and second primaries.
- **Commonest sites** in India and in the West.
- **Tongue cancer:** site, spread, depth of invasion, referred otalgia and treatment, including the commando operation.
- **Mandible:** marginal versus segmental resection.
- **HPV-related oropharyngeal cancer** and the cystic neck node.
- **Management of the N0 neck.**
- **Differential diagnosis of a one-sided enlarged tonsil.**

## 8.2 Common pitfalls

COMMON PITFALLS

- Treating a persistent ulcer with mouthwashes rather than a biopsy.
- Judging a tongue base cancer by looking without feeling.
- Excising a cystic neck lump as a branchial cyst in an adult without looking for a primary.
- Irradiating a mandible already invaded by tumour.

Written for this app from Oxford Textbook of Otolaryngology ch. 56, Oxford Specialist Handbook ch. 6, Dhingra ch. 44, 53 and K. J. Lee ch. 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

## Tonsillar Carcinoma

### IMAGING FINDINGS

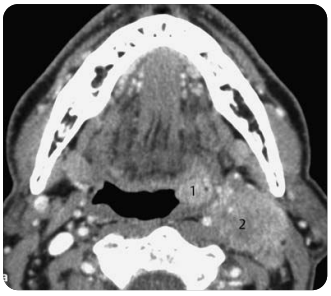
- Asymmetric enhancing tonsillar mass, sometimes small
- Spread to soft palate, tongue base and parapharyngeal space
- Cystic or necrotic level II nodes (HPV)

### TOP DIFFERENTIALS

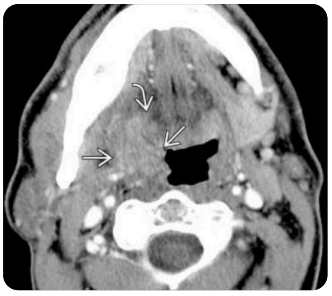
- Tonsillar lymphoid hyperplasia
- Lymphoma
- Peritonsillar abscess

### CHECKLIST

- Medial pterygoid, mandible and carotid
- Both sides of the neck




CT with intravenous contrast enhancement, axial. In the left tonsillar fossa, a small and enhancing tumor is visible (1), with enlarged level II lymph…



Axial CECT in alcoholic smoker presenting with multiple right neck masses shows an enlarged right palatine tonsil mass…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 492.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

## Oropharyngeal Tumor, Base of Tongue

### IMAGING FINDINGS

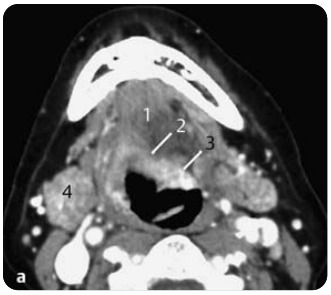
- Asymmetric fullness of the lingual tonsil with enhancement
- Spread to vallecula, pre-epiglottic space and floor of mouth
- Level II nodes, often bilateral

### TOP DIFFERENTIALS

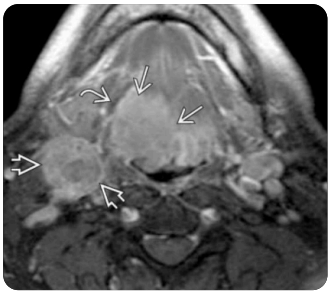
- Lingual tonsil lymphoid hyperplasia
- Lymphoma
- Thyroglossal or vallecular cyst

### CHECKLIST

- Crossing the midline and neurovascular bundles
- Pre-epiglottic and laryngeal invasion




CT with intravenous contrast enhancement, axial. Axial slice at the level of the mandible and floor of the mouth (1). At clinical evaluation, an…



Axial T1WI C+ FS MR in a 58-year-old alcoholic presenting with persistent right neck mass after dental procedure demonstrates enlarged,…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 488.

ORAL CAVITY & TONGUE · AFTER SURGERY

## Squamous cell carcinoma: oral cavity, oropharynx and larynx

### IMAGING FINDINGS

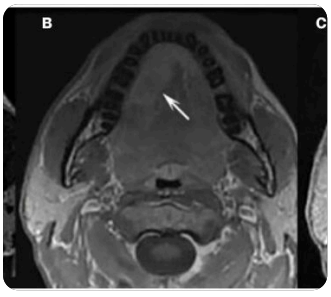
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

### TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

### CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT




Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



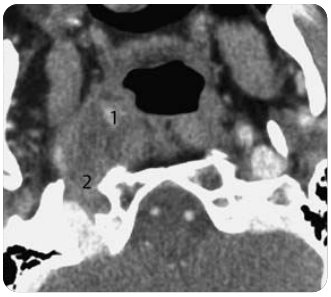
Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.

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# Multiple Synchronous Malignancies

- Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and spread of each tumor to the contralateral side.
- Lymphomas (non-Hodgkin) initially may present as unilateral lymphadenopathy and ulcerative lesions.
- Evaluation of systemic disease and lymph nodes in other regions.
- Keep in mind that (micro)metastases may be radiographically occult.




CT with intravenous contrast enhancement, axial.A lesion with malignant characteristics is seen in the right tonsillar fossa (1). The…